CHAPTER 1

INTRODUCTION

Spirulina (Arthrospira platensis) is a photosynthetic, filamentous blue–green microalga that is commonly known as a "superfood" due to its extremely high protein content, vitamins, essential amino acids, minerals, and bioactive pigments like phycocyanin and chlorophyll-a. It has extensive application as a food supplement, functional food ingredient, and natural source of pigments that exhibit antioxidant and immunomodulatory activities. Its high growth rate, high alkalinity tolerance, and capacity for carbon fixation justify it as a promising candidate for the production of sustainable biomass.

Traditional commercial Spirulina production normally utilizes artificial nutrient media and very controlled photobioreactors or raceway ponds. Although the former provides high yields and constant quality, they are costly to run and environmental problems related to the production of artificial nutrients. In recent years, interest has been on the development of cost-efficient and sustainable cultivation processes utilizing agro-industrial or domestic waste streams as substitute sources of nutrients. These methods not only lower the cost of production but also value organic waste, in further of circular economy and environmental sustainability.

Spirulina was grown in this research using nutrient-rich food waste extracts—banana peel extract, aquafaba (legume cooking water), and peanut meal extract—in place of traditional synthetic nutrients. A high-quality mother culture procured from Viridia Biotech Company, Karur was utilized as the inoculum for maintaining a good starting population. The

cultivation process was carried out on the terrace of Kongu Engineering College, utilizing natural sunlight and ambient conditions to similar a low-cost, semi-outdoor production system.

Response Surface Methodology (RSM) was used to optimize the ratio of the three food-waste components in order to maximize Spirulina biomass production and enhance its nutritional value. The produced Spirulina was subsequently characterized for proximate composition (carbohydrates, protein, fat), pigments (chlorophyll-a, β-carotene), minerals, color, antioxidant capacity and energy value and compared to commercially available Spirulina. This project shows the viability of cultivating nutrient-rich Spirulina in low-cost, sustainable media under easy growth conditions. It also supplies information regarding the effects of medium composition and environmental conditions on the biochemical profile of Spirulina and hence is useful for designing large-scale eco-friendly Spirulina production systems.

1.1 OBJECTIVES

- To identify and collect the suitable nutrient rich sources from the food waste.
- To optimize the composition of the nutritional medium by RSM software.
- To evaluate the growth performance and nutritional profile of harvested spirulina.
- To compare the nutritional profile of cultivated spirulina (by optimized medium) with spirulina grown in commercial medium.

CHAPTER 2

REVIEW OF LITERATURE

Fontaine *et al.* (1944) conducted one of the first comprehensive studies on the nutritional and functional characteristics of peanut meal. Their study examined proximate composition, with emphasis on protein, lipid, carbohydrate, and fiber content. They emphasized peanut meal's high protein content, making it an excellent candidate to replace traditional protein sources, especially given its low cost. Functional properties of water absorption and emulsifying capacity were recognized and suggested its application in food and feed formulation. In an era where plant-based proteins did not receive significant attention for human foods, this research laid down much-needed foundations for subsequent investigation of oilseed by-products as functional food ingredients and illustrated the application of peanut meal to enhance dietary protein consumption in resource-poor environments.

Belay *et al*. (1993) had given a pioneering review of the nutritional composition and therapeutic value of Spirulina. They recognized Spirulina as an outstanding source of protein (55–70%) with all the essential amino acids, and valuable fatty acids like gamma-linolenic acid (GLA). They also emphasized its richness in vitamins (provitamin A and B-complex vitamins), minerals (iron, magnesium, calcium), and other special pigments like phycocyanin and carotenoids. In addition to nutritional values, Spirulina exhibited antioxidant, anti-inflammatory, and immunomodulatory activities. The authors highlighted its practicality against malnutrition and micronutrient deficiency in developing nations with ease of cultivation and minimal

environmental footprint. Their research established Spirulina as a functional food and nutraceutical, shaping decades of follow-up research and uses in health supplements.

Cohen *et al.* (1997) provided a detailed chemical characterization of Spirulina, describing its key macronutrient and micronutrient composition. Spirulina was characterized as having high-quality protein with balanced amino acids, moderate polysaccharides (carbohydrates), and essential fatty acids. Of particular interest were its bioactive compounds such as phycobiliproteins, phycocyanin, and carotenoids, which have antioxidant and therapeutic activities. Cohen's research also highlighted Spirulina's pharmaceutical potential, referring to its antiviral, anticancer, and immunostimulant activities. In addition, the research highlighted the promise of Spirulina pigments as natural substitutes for man-made food colorants, hence, a desirable option for the food industry. This review took the previous studies further by elucidating the biochemical bases for Spirulina's nutritional and health-beneficial actions, hence, providing critical information for nutraceutical formulation.

Falquet & Hurni (1997) explored Spirulina's nutritional attributes in relation to its contribution toward correcting malnutrition globally. Their analysis reinforced that Spirulina consists of approximately 60–70% protein, with a high content of essential amino acids, and is rich in amounts of vitamins like beta-carotene, vitamin B12, and minerals like iron. They also highlighted the availability of GLA, an essential fatty acid that is useful for cardiovascular and immune system well-being. Spirulina's ease of digestion, as well as its capacity to produce under conditions of high resource efficiency, was emphasized as a strength in the Fig.ht against hunger in less developed nations. Their research placed Spirulina not only as a health supplement but also as a sustainable remedy for protein-energy malnutrition, with immediate socio-economic relevance for world food security.

Sanchez *et al*. (2003) presented a comprehensive review of Spirulina (Arthrospira) as a food-grade microorganism, covering its cultivation practices, safety profile, nutritional composition, and possible applications. They noted Spirulina's safety as a food organism and explained its acceptability for use in fortified foods, beverages, and dietary supplements. The

review underscored Spirulina's protein content, essential amino acids, and distinctive bioactive pigments such as phycocyanin, which exhibit antioxidant and immunomodulatory effects. They concluded that Spirulina possesses great potential in providing human nutrition and food security, particularly in areas with dietary deficiencies.

Batal *et al*. (2005) examined the nutritional profile of peanut meal and its protein content, as well as its functional characteristics. They reported high-quality protein content of peanut meal, along with good emulsifying and foaming properties, water- and oil-absorbing capabilities, and good amino acid balance. These functional properties render peanut meal ideal not just for animal feed but also as a functional food ingredient in human foods, where it can enhance texture, stability, and nutritional content. Their research widened the scope of peanut meal applications, noting it as an underdeveloped by-product with significant potential for functional food development.

Ali & Saleh (2012) presented an in-depth review of Spirulina, including cultivation, nutritional value, and industrial use. They emphasized Spirulina's quality of protein, essential fatty acids, vitamins, and antioxidant pigments. Their review also touched upon Spirulina's pharmaceutical and nutraceutical potential, including its use to treat chronic diseases like diabetes, hypertension, and hyperlipidemia. Spirulina's application as a natural coloring agent in foods because it contains phycocyanin was also mentioned. The research placed Spirulina as a multipurpose material for pharmaceuticals and food industries, making it one of the most viable microalgae for commercial uses.

Saranraj & Sivasakthi (2014) summarized Spirulina platensis as "food for the future" highlighting its high content of protein, vitamins, and pigments. They were interested in the health-promoting aspects of Spirulina, including its antioxidant and anti-inflammatory effects, cholesterol reduction, and immunoenhancing capability. Their research also emphasized Spirulina's ability to lower risks of lifestyle disorders like cardiovascular disease and obesity. They concluded that Spirulina is promising for large-scale integration into nutraceuticals and fortified foods with the potential to provide not only nutrition but also therapeutic effects.

Rosario & Josephine (2015) also carried out an extensive review of Spirulina's mineral content, making it a rich mineral source of the vital mineral's calcium, iron, magnesium, and potassium. According to their results, Spirulina has the potential to act as a natural mineral supplement against micronutrient deficiency, especially iron-deficiency anemia. The writers highlighted Spirulina's significance in food supplementation programs and its suitability for incorporation into functional foods aimed at treating mineral deficiency in humans.

Merz & Main (2017) investigated the bioproduction of microalgae with particular emphasis on Spirulina, outlining its uses in food, feed, nutraceuticals, and polymers. They elaborated on recent developments in large-scale production, harvesting, and processing technology, improving output and cost-benefit. Spirulina was also presented as a renewable and sustainable resource, supplying proteins, pigments, and bioactive compounds. Notably, they mentioned its applications beyond food, including biodegradable polymers, which is giving impetus to the bioeconomy and circular economy patterns.

Soni *et al.* (2017) conducted a review of Spirulina's evolution from production to industrialization as a food supplement. Their review highlighted Spirulina's bioactive molecules such as phycocyanin, polysaccharides, and carotenoids, with their focus on promoting health in terms of antioxidant, immune-modulatory, and anti-inflammatory activities. They also discussed the addition of Spirulina into processed foods such as snacks, drinks, and supplements, showing its market viability. The review further solidified Spirulina's reputation as a nutrient-dense food material and as a functional food material.

Liestianty *et al.* (2019) examined the nutritional profile of Spirulina sp. and confirmed its status as a "superfood." They reaffirmed Spirulina's very high protein value, balanced amino acid profile, and high micronutrient value, including vitamins and minerals. They also underscored its bioactive compounds, which endow it with antioxidant and therapeutic properties. The study advised the incorporation of Spirulina into health foods and supplements to combat malnutrition and lifestyle diseases such as obesity, diabetes, and oxidative stress.

Alsalman *et al*. (2020) researched aquafaba, the watery liquid from cooked legumes, assessing its functional and antinutritional characteristics. They optimized its use as an egg replacement with respect to foaming ability, emulsification, and protein quality. While emphasizing its prospect in vegan and allergen-free foods, they also tackled shortfalls due to antinutritional factors. Their results showed aquafaba's potential in bakery and confectionery applications as a sustainable substitute for animal-derived materials.

Grosshagauer *et al*. (2020) evaluated the nutritional, bioactive, and functional attributes of Spirulina. They emphasized its quality protein, antioxidant substances, and pigments like phycocyanin, which bring nutritional value as well as potential beneficial health effects. Spirulina was also acknowledged as a sustainable food material with extensive uses in dietary supplements, functional foods, and health-enhancing products. The review reaffirmed Spirulina's status as a "green superfood" for contemporary food systems.

AlFadhly *et al.* (2022) discussed technological advances and innovations in food uses of Spirulina. They discussed novel processing methods, added to beverages, snacks, and supplements, and its medicinal application in alleviating oxidative stress, inflammation, and metabolic disorders. Their review indicated the versatility of Spirulina across different food forms and pointed toward the potential of large-scale industrial application as a multifunctional ingredient.

Hikal *et al*. (2022) investigated banana peels as an underexploited by-product high in fiber, polyphenols, and antioxidants. Their overview highlighted the nutritional and bioactive content of banana peels as proposing their valorization to functional food ingredients, dietary supplements, and natural antioxidants. They also outlined the environmental and economic advantages of banana peel utilization within circular economy and waste-reduction approaches.

Ansari *et al*. (2023) built on banana peel research by outlining its nutritional and bioactive content, such as dietary fibers, flavonoids, and minerals. They suggested banana peel's application in functional foods, nutraceuticals, and drug formulations for disease prevention and sustainable food processing. Their review also identified the potential of banana peel valorization to mitigate food waste and enhance resource efficiency.

Hariharan *et al*. (2023) investigated future processing technologies to recover functional proteins from peanut meal. They elaborated on enzymatic hydrolysis, ultrafiltration, and other novel approaches to increase protein solubility and functionality. Their results showed that peanut meal proteins have potential in enhancing texture, emulsification, and nutritional value in food systems. The review pointed out the peanut meal bio valorization as a viable approach to sustainable food industry protein production.

Bishnoi *et al*. (2023) evaluated the valorization value of banana peels, focusing on their antioxidant, dietary fiber, and bioactive compound contents. They foregrounded the health benefits of banana peels in formulating nutraceuticals, functional food additives, and therapeutic drugs. The research also underlined the environmental viability of the use of banana peels, advocating for its application in circular economy strategies and waste mitigation.

Satta *et al.* (2023) summarized metabolic engineering technologies for photosynthetic bioproduction in cyanobacteria, especially Spirulina. They summarized recent genetic manipulations to enhance bioactive compound yields, pigments, and proteins. Their review focused on the industrial promise of engineered Spirulina strains for commercial applications in functional foods, drugs, and bio-based materials. This article set the direction of Spirulina biotechnology in the future as an instrument for sustainable production of valuable compounds.

CHAPTER 3

MATERIALS AND METHODOLOGY

3.1 RAW MATERIALS

Raw materials utilized in this work comprised both the nutrient sources from food waste and Spirulina inoculum. Banana peel extract, aquafaba (liquid from boiled beans), and peanut meal extract were chosen as inexpensive and nutrient-dense substrates to serve as substitutes for traditional synthetic media. They offer carbohydrates, proteins, and minerals required to cultivate Spirulina. Clean water was employed as the base medium for the preparation of the nutritive solutions. A good-quality Spirulina mother culture procured from Viridia Biotech Company, Karur, was used as the inoculum for having a healthy and uniform starting population. All the raw materials were procured fresh, handled properly, and stored under hygienic conditions to preserve their nutritional integrity before use in experimental trials.

3.2 PROCUREMENT OF MOTHER CULTURE

The Spirulina mother culture used in this study was sourced from Viridia Biotech Company, Karur, Tamil Nadu, which specializes in producing high-quality microalgal cultures. It was chosen to ensure a pure, contaminant-free, and actively growing inoculum suitable for the experiments. The culture was transported under refrigerated and sterile conditions to maintain its viability and prevent contamination during transit. On arrival, it was checked for its

color, odor, and uniformity to confirm its healthy state before being introduced into the prepared nutrient medium.

3.3 LOCATION AND CULTIVATION CONDITIONS

The Spirulina culture experiments were conducted on the rooftop of Kongu Engineering College in order to make use of sunlight and open-air conditions, thus resembling a low-cost, semi-outdoor system of production. The culture was maintained in 1000 ml capacity glass beakers with the optimized nutritive medium at a 1:10 (nutrient medium: water) ratio. The medium's initial pH was regulated between 8.5 and 10, whereas the temperature on the terrace varied from 30–38 °C, which is a favorable condition for the growth of algae. Cultures were aerated or stirred gently every now and then to avoid settling and to get all the cells equally exposed to nutrients and light. These growth conditions favored visible biomass increment within 8–10 days, which aligns with the vegetative growth stage of Spirulina under such environmental conditions.

3.4 MEDIA COLLECTION

The nutrient medium for Spirulina was prepared using extracts from selected food waste materials such as banana peels, water from boiled legumes (aquafaba), and peanut meal extract. These materials were collected fresh from industrial food waste to make sure they were clean and nutrient-rich. After collection, the extracts were filtered to remove larger particles and then measured carefully to get the right formulation. Clean water was used as the base, and for every 100 ml of water, 10 ml of the prepared nutrient extract was added as per the optimized composition. To maintain freshness and reduce the risk of contamination, the media were always prepared just before inoculation.

3.5 EXPERIMENTAL SETUP

The experiment was carried out to study how Spirulina grows in different nutrient mixtures made from food waste extracts. The cultivation was done in 1-liter glass beakers placed on the college terrace, where they received natural sunlight and temperature conditions. Each beaker was filled with the nutrient medium mixed with water in a 1:10 ratio, and then inoculated with a fixed amount of Spirulina mother culture. The growth environment was maintained at a pH between 8.5 and 10 and a temperature of 30–38 °C throughout the study.

To optimize the medium composition, Response Surface Methodology (RSM) software was used. A total of 11 experimental trials were carried out with varying levels of banana peel extract, aquafaba, and peanut meal extract to identify the best combination for maximum Spirulina biomass. The cultures were gently shaken at intervals during the 8–10-day growth period to ensure proper mixing of nutrients and uniform light exposure. At the end of each trial, the Spirulina was filtered, dried using a tray dryer at 40–50 °C, and processed into pellets.

3.6 MEDIA PREPARATION

The culture medium for the growth of Spirulina was newly prepared before each run of experiments using food waste–derived constituents. Banana peel extract, aquafaba derived from boiled legumes, and peanut meal extract were extracted, filtered, and weighed according to the optimized proportions established by Response Surface Methodology (RSM). These supernatants were then mixed with sterile water in the ratio of 10 ml nutritive medium per 100 ml water, providing a consistent and reproducible cultivation environment for Spirulina. The medium was adjusted to have a pH between 8.5–10, which is ideal for the growth of Spirulina. All media were produced under sanitary conditions and employed promptly to maintain their nutrient stability and reduce microbial contamination prior to inoculation of the Spirulina mother culture.

3.7 METHODOLOGY FOR THE CULTIVATION OF SPIRUNILA

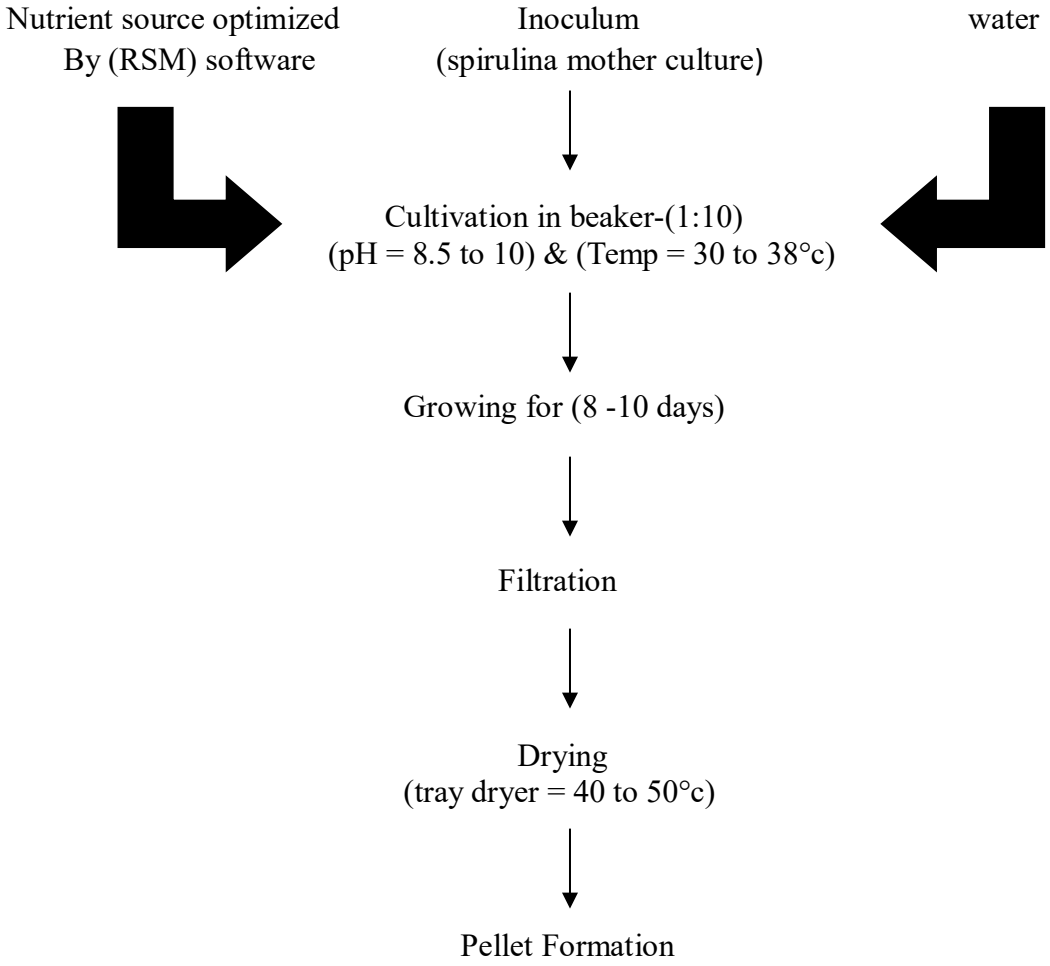


Fig. 3.1 Flowsheet of spirulina cultivation

3.8 OPTIMIZATION OF COMPOSITIONAL CHANGE IN NUTRITIONAL MEDIUM

Response Surface Methodology (RSM) was utilized to optimize the formulation of the Spirulina growth medium with food waste–derived nutrient sources. Banana peel extract, aquafaba, and peanut meal extract, the three chosen variables, were taken as independent factors, whereas the Spirulina biomass yield and protein was used as the dependent response. 11 trial

experiments were planned to investigate the influence of these variables at varying concentration levels and their interactions on growth performance. The cultures of each trial were maintained in controlled pH (8.5–10) and temperature (30–38 °C) conditions with 8–10 days of growth. Results from the trials were statistically processed through the RSM model to contour plots, predicting the best nutrient formulation. Optimization was thus used to develop a low-cost, nutrient-dense medium that could sustain higher Spirulina biomass yields than standard culture media. The upper and lower limits were demonstrated in the Table:3.1

Table 3.1 Upper and lower limits of the mixture design

	Name	Low	High
A[Mixture]	Banana Peel Extract(ml)	2ml	6ml
B[Mixture]	Aquafaba(ml)	2ml	6ml
C[Mixture]	Peanut Meal Extract(ml)	2ml	6ml

3.9 ESTIMATION OF ASH CONTENT

The ash content of the developed spirulina has been determined by weighing accurately 10 g of formulated feed into a weighed crucible. It is incinerated in the muffle. The Petri dish was kept in a muffle furnace maintained at 550-600 °C until light grey ash results or to a constant weight then cooled in a desiccator and weighed. The ash was found out from the Equation (3.1).

$$Total\ Ash\ Content = (W_2\text{-}W) \div (W_1\text{-}W) \times 100 \qquad (3.1)$$

Where:

W_1 = Weight of the crucible with the dried material taken for the test in (g)

W_2 = Weight of the crucible with the ash (g)

W = Weight of the empty crucible (g)

3.10 ESTIMATION OF CARBOHYDRATES

To extract sugars from starch, homogenize the sample (spirulina) in hot 80% ethanol centrifuge, and wash it multiple times until it no longer shows color. Dry the residue in a water bath, and then add distilled water and 52 % perchloric acid to the residue. Centrifuge for 20 min at 0 °C, repeating with fresh perchloric acid. Combine the extracts, pipette 0.1 ml into tubes, fill 1.0 ml with pure water, add 4.0 ml of anthrone reagent, and boil in a water bath for 8 min. Cool and measure the color at 620 nm.

3.11 ESTIMATION OF WATER ACTIVITY

Water activity is the measure of amount of free water present in the food for microbial contamination. The water activity of developed feed was measured using Aqualab digital water activity meter.

3.12 ESTIMATION OF PROTEIN

The protein concentration was determined by using Lowry's Method, working standard is prepared using Bovine serum albumin. Working standard is pipetted in a series of test tube from 0.2-1.0 and water is added to make up the volume to 1 ml in all the test tube. A tube with 1ml water serves as blank. 5 ml of a reagent which is a mixture of 1ml of sodium

potassium tartrate and copper sulphate and 50 ml of 0.1 N NaOH is added to each testtube.0.5 ml of Folin's reagent is added to each test-tube and mixed well and kept in dark for 30mins and the absorbance is measured in 660 nm.

3.13 DETERMINATION OF FAT CONTENT

The sample (spirulina) is placed in a thimble; once the flask is heated, the solvent is evaporated and moved up to the condenser, where it is converted into a liquid and collected into the extraction chamber containing the sample. When the solvent passes through the sample, it extracts the fats and carries them into the flask. This extraction process typically lasts several hours (1-2 hr). After completion of the extraction, the solvent is evaporated, and the mass of lipid remaining is measured and used for analysis using the Equation (3.2).

$$Crude\ Fat\ (\%) = \frac{W1 - W2}{W} \times 100 \qquad (3.2)$$

Where;

W = Weight of sample in gram

W_1 = Weight of sample in gram +Weight of Soxhlet flask in gram

W_2- Weight of Soxhlet flask in gram

3.14 DETERMINATION OF CHLOROPHYLL a

5 g of spirulina is mixed with 25 ml of 95 % of ethanol and boiled at 78 ℃ for five minutes. After boiling the sample is at dark room for 24 hours. The sample is mixed thoroughly and centrifuged for 10 mins at 4500 rpm. The supernatant is collected and absorbance was

measured in 663 nm and 645 nm with 95 % ethanol serving as blank and the obtained value is get substituted is the Equation (3.3)

$$Chlorophyll\ a\ [\frac{\text{mg}}{\text{g}}] = (12.7 * \text{O.D}\ 663) - (2.69 * \text{O.D}\ 645) \qquad (3.3)$$

3.15 DETERMINATION OF ß-CAROTENE

5 g of sample is centrifuged with acetone and the supernatant is collected continuously until the residue turns colourless. Pooled acetone is transferred to a separating funnel containing 20ml of petroleum ether. 20 ml of 5 % sodium sulphate is added and shake gently. upper layer is removed and mixed with 10 g anhydrous sodium sulphate to remove any moisture and kept aside for 30 mins. Decant the petroleum ether into 100 ml volumetric flask and made up with petroleum ether. The absorbance was measured at 453 nm using petroleum ether as blank, and the obtained value is get substituted is the Equation (3.4)

$$ß\ Carotene\ = \frac{absorbance\ of\ sample * total\ volume * 100}{0.2592.weight\ of\ sample.1000} \quad mg \qquad (3.4)$$

3.16 DETERMINATION OF MINERALS

The determination of sodium, calcium, potassium is done by using a flame photometer. A 5g sample is taken and dried in muffle furnace at 550 ℃ for 5 hr. The ash obtained is dissolved in 5ml HNO_3/HCI/H_20 at 1:2:3 ratio, then heated on hot plate until the brown fumes disappear. To the remaining residue, 5 ml of distilled water is added and heated till colourless solution is obtained. The obtained mineral solution is filtered and transferred to 100 ml volumetric flask, make up to the mark with distilled water. Hence the sample solution is prepared. The prepared sample solution is tested by dipping in the nebulizer. The sample will

aspire and colour of the flame will change depending on the sample. The readings are displayed and compared with the standard. The value of mineral content present in the sample is obtained.

3.17 DETERMINATION OF COLOUR

Color analysis was carried out using a CM-5 spectrophotometer. A small quantity of the spirulina powder was placed in a clean petri dish and positioned on the measurement port of the instrument. The device was calibrated using standard white and black tiles prior to analysis. The sample was measured under standard illuminant D65 and a 10° observer angle. The resulting L*, a*, and b* values, representing lightness, red-green, and yellow-blue components respectively, were recorded directly from the instrument display from the Equation (3.5).

$$\Delta E^*_{ab} = \sqrt{\Delta L^{*2} + \Delta a^{*2} + \Delta b^{*2}} \qquad (3.5)$$

3.18 DETERMINATION OF ANTIOXIDANTS

The antioxidant activity of the spirulina was assessed using the DPPH (2,2-diphenyl-1-picrylhydrazyl) radical scavenging method, which evaluates the ability of bioactive compounds to neutralize free radicals by donating hydrogen atoms or electrons. In this process, a food sample's antioxidant potential is determined through a series of reactions and measurements. First, a DPPH solution is prepared in ethanol and protected from light to preserve its radical stability. A known volume of this DPPH solution is then mixed with a measured quantity of the food extract containing antioxidant compounds. The mixture is incubated in the dark at room temperature, allowing the antioxidants present in the sample to interact with and reduce the DPPH radicals. This reaction causes a visible color change from deep violet to pale yellow, indicating radical neutralization. Following the incubation period, the absorbance of both the control solution (DPPH without sample) and the sample solution (DPPH with food extract) is measured spectrophotometrically at a wavelength of 517 nm. The difference in absorbance values is used to calculate the percentage of DPPH radicals scavenged by the

sample. These steps are part of a standardized method for assessing the antioxidant activity of food samples. Equation (3.6) is used to calculate the DPPH scavenging percentage.

$$DPPH\ Scavenging\ Activity(\%) = \frac{Working\ standard - Sample}{Working\ standard} \times 100 \qquad (3.6)$$

3.19 DETERMINATION OF ENERGY VALUE

The calorific value of the spirulina was measured in a bomb calorimeter. The combined lid of the calorimeter was first put together. The calorimeter vessel containing a groove was placed in the slot at the bottom of the water jacket into which the triangular support was inserted. The inlet cover on the jacket was taken out by loosening the screws, and around 16–17 L of water was filled into the jacket up to 5–6 cm from the top before putting the cover back. The bomb support was subsequently placed within the calorimeter vessel such that it was held firm against the "U" shaped bracket and could not shift. The small pin mounted at the bottom of the vessel calorimeter was inserted into the slot on the triangular support, then the vessel was settled into the water jacket. The oxygen charging system was assembled and the bomb lid placed on the provided stand with the pressure gauge. Ignition electrodes were mounted and a 6 cm ignition wire tightly stretched between both electrodes.

A pellet of the spirulina having a weight less than 1.10 g (or halved or one-third the weight if the calorific value was higher than 10,000 Cal/g) was made and positioned in the crucible, which was subsequently inserted into the crucible holder ring. A thread was knotted in the middle of the ignition wire such that its end was in contact with the pellet. About 1–2 ml of distilled water was pipetted into the bomb, and the bomb was thereafter sealed by hand and oxygen-charged. The bomb was then hung in the calorimeter vessel by virtue of the provided hook, with bottom support. Electric connections were established by pushing the plug wires from the bomb terminals through the slot on the combined lid into the sockets at the rear of the firing unit. Measured water was poured into the calorimeter vessel, and the combined lid was

refitted such that the small pin on the jacket registered with the slot on the lid. The stirrer was mounted on the pole available on the jacket and its propeller submerged in the water via the lid holes. The temperature sensor plug was pushed inside the combined lid into its connector, the mains cord of the digital firing unit to 220 V (AC), 50 Hz connected and powered on, and the plug of the stirrer inserted and powered on.

At the end of 2–3 minutes, the firing unit switch was shifted from "Temp. Initial" to "Temp. Rise" mode and the reading on the display was set to 0.00 using the "zero adj." knob. The "fire" button was now pressed to burn the sample. The temperature was observed as it increased, levelled off, and started falling, with the top temperature noted as the last rise in temperature (T). After measurement was finished, the firing unit and stirrer were turned off, the stirrer and sensor were removed, and the combined lid was opened. Lead from the bomb was removed, and the bomb was lifted from the vessel by the hook. Pressure in the bomb was released through the gas release pin before disassembly. The interior of the bomb, including electrodes and crucible, was washed with water and a soft cloth or with compressed air dried. Equation (3.7) is employed to determine the calorific value of food sample.

$$Calorific\ value\ of\ the\ sample = \frac{\left(T \times W - (Cvt + Cvw)\right)}{M} \qquad (3.7)$$

Where;

T = Temperature Rise measured from the Instrument

W = Water Equivalence (2332 Cal/ºC- Value Standardized using benzoic Acid)

Cvt = Length of the thread * 2.1 Cal/ºC

Cvw = Length of the wire * 2.33 Cal/ºC

M = Mass of the Sample (g)

3.20 ESTIMATION OF CRUDE FIBER

The crude fiber content of the developed spirulina was determined by subjecting the sample to a series of chemical treatments to remove all soluble and easily digestible components. 1 g of the spirulina were extracted. The developed spirulina was then boiled with 200 ml of sulfuric acid for 30 min, with bumping chips used to prevent splashing. It was filtered through muslin and washed with boiling water until the washings were no longer acidic. The residue was boiled with 200 ml of sodium hydroxide solution for 30 min and then filtered through muslin cloth again. The residue was removed and transferred to a crucible. It was dried for 2 hr at 100 °C in hot air oven. The dish was cooled in a desiccator and weighed. Afterward, it was ignited for 2 hr at 550 °C, cooled in a desiccator, and reweighed. The crude fiber content was determined using Equation (3.8).

$$\text{Crude fiber Content} = (W_2\text{-}W_3)/W_1 * 100 \qquad (3.8)$$

Where:

W_1=weight of sample (g)

W_2=weight of dried residue with crucible (g)

W_3=weight of ash with crucible (g)

3.21 ESTIMATION OF MOISTURE CONTENT

The moisture content of the developed animal feed has been determined by AOAC method, weighing accurately about 5 g of the feed in the petri dish, previously dried in the oven at 105˚C and weighed. The dish is placed in the oven maintained at 130˚C for 1 hour and cooled in the desiccator and weighed until constant weight are reached. The moisture content was found out from the Equation (3.9).

$$Moisture\ Content(\%) = \frac{M1-M2}{M1-M} \qquad (3.9)$$

Where;

M_1 = Weight of the dish with the material before drying (g)

M_2 = Weight of the dish with the material after drying to constant weight (g)

M = Weight of the empty dish (g)

CHAPTER 4

RESULT AND DISCUSSION

Spirulina was successfully grown using food waste-based nutrient sources in 1000 ml beakers placed on the college terrace under natural light and environmental conditions. A clear increase in biomass was observed within 8–10 days, showing healthy cell division and photosynthesis. The eleven RSM trials highlighted that varying the levels of banana peel extract, aquafaba, and peanut meal extract had a direct effect on Spirulina growth. Balanced proportions of these three inputs supported higher cell densities, whereas very low or very high concentrations slowed down growth, confirming the importance of optimizing nutrient ratios.

The statistical analysis of the RSM results showed a good model fit with strong interactions among the three nutrient sources, which helped in accurately predicting the best medium composition. Spirulina cultivated in the optimized medium gave a noticeably higher biomass compared to the control. The harvested biomass retained its bright green color and firm texture after filtration and tray drying at 40–50 °C, indicating good quality. Overall, the findings suggest that low-cost media made from food waste can effectively replace expensive synthetic nutrients, adding value to waste while maintaining high Spirulina productivity in simple semi-outdoor conditions.

4.1 OPTIMIZATION OF THE NUTRITIONAL MEDIUM

Table 4.1 Responses for RSM Trials of the spirulina cultivation

Run	Component 1 Banana Peel Extract (ml)	Component 2 Aquafaba (ml)	Component 3 Peanut Meal Extract (ml)	Response 1 Biomass (g)	Response 2 Protein (%)
1	2	4	4	7.25	56.98
2	6	2	2	4.47	41.21
3	2	2	6	9.71	68.86
4	4	4	2	4.29	40.76
5	3.33333	3.33333	3.33333	7.63	55.31
6	2.66667	2.66667	4.66667	7.95	58.74
7	4	2	4	6.55	54.06
8	2.66667	4.66667	2.66667	5.22	43.96
9	6	2	2	4.38	41.64
10	4.66667	2.66667	2.66667	5.42	46.88
11	2	6	2	5.54	46.12

The Response Surface Methodology (RSM) experimental plan with eleven trials compared the influence of three food-waste-derived ingredients—banana peel extract, aquafaba and peanut meal extract—on Spirulina development. The outcome indicated extensive variance in biomass yield (4.29–9.71 g) and protein content (40.76–68.86%) based on the composition of the medium. Maximum performance was observed in Run 3 with 2 ml banana peel extract, 2 ml aquafaba and 6 ml peanut meal extract, producing maximum biomass (9.71 g) and protein (68.86%). The central composite point (Run 3) with the proportions of all components proved

satisfactory (9.71 g biomass and 68.86% protein), reflecting that higher nutrient levels greatly support Spirulina growth from the Table 4.1.

Conversely, experiments carried out with extremely high or extremely low levels of a single component, for example, high banana peel extract with minimum aquafaba or peanut meal extract, yielded relatively lower levels of biomass and proteins. This trend suggests that it is crucial to have an optimal combination of all three sources of nutrients instead of a surplus in one component. The RSM model successfully reflected these interactions and identified the optimum combination of parameters for achieving maximum Spirulina biomass and protein content employing low-cost, environmentally friendly substrates.

4.1.1 Effects on composition of nutritional medium on Biomass

Table 4.2 ANOVA for linear model of Biomass

Source	Sum of Squares	df	Mean Square	F-value	p-value	
Model	28.2	5	5.64	11.7	0.0087	significant
[(1)]Linear Mixture	28.15	2	14.07	29.2	0.0017	
AB	0.044	1	0.044	0.0913	0.7747	
AC	0.001	1	0.001	0.0021	0.965	
BC	0.0051	1	0.0051	0.0106	0.9219	
Residual	2.41	5	0.4819			
Lack of Fit	2.41	4	0.6014	148.49	0.0615	not significant
Pure Error	0.004	1	0.004			
Cor Total	30.61	10				

Table 4.3 Fit statistics of the of Biomass

Std. Dev.	0.6942	**R^2**	0.9213
Mean	6.22	**Adjusted R^2**	0.8425
C.V. %	11.16	**Predicted R^2**	0.1564
		Adeq Precision	10.15

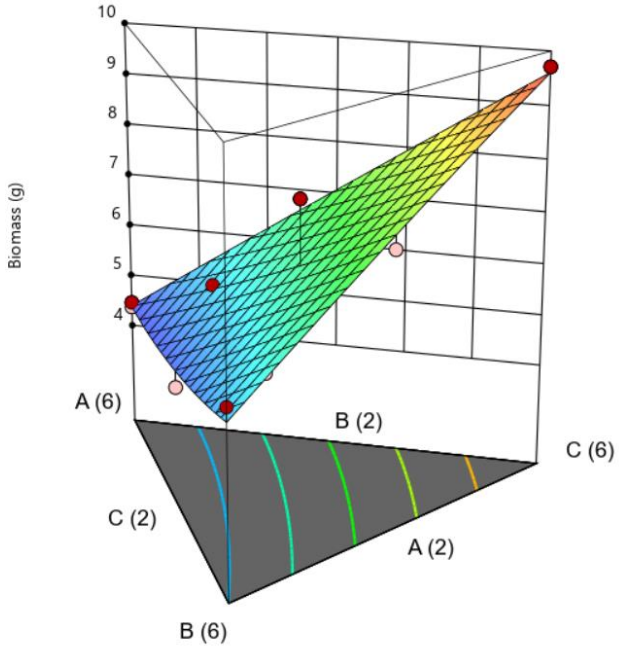


Fig. 4.1 Effects on composition of nutritional medium on Biomass

The ANOVA results for the linear model of Biomass given in Table 4.2 showed that the overall model was statistically significant, with an F-value of 11.7 and a p-value of 0.0087, confirming that the selected nutrient variables had a strong effect on Spirulina biomass production. Among the terms, the linear mixture effect was highly significant (F-value 29.2, p = 0.0017), while the interaction terms AB, AC, and BC were not significant (p-values 0.7747, 0.9655, and 0.9219), suggesting that the main effects of the nutrients were more important than their combined interactions. The lack of fit was not significant (p = 0.0615). Model fit statistics further supported this, with a high R^2 value of 0.9213 and an adjusted R^2 of 0.8425, though the

predicted R^2 (0.1564) was comparatively low. The average response was 6.22 g, with a standard deviation of 0.6942 and a coefficient of variation of 11.16%, indicating reasonable consistency. An adequate precision value of 10.15 confirmed a strong signal-to-noise ratio, showing that the model could effectively navigate the design space Table 4.3.

4.1.2 Effects on composition of nutritional medium on Protein

Table 4.4 ANOVA for linear model of Protein

Source	Sum of Squares	df	Mean Square	F-value	p-value	
Model	779.8	5	155.96	20.72	0.0024	significant
[(1)]Linear Mixture	776.5	2	388.25	51.58	0.0005	
AB	3.21	1	3.21	0.427	0.5423	
AC	0.1564	1	0.1564	0.0208	0.891	
BC	0.0235	1	0.0235	0.0031	0.9576	
Residual	37.64	5	7.53			
Lack of Fit	37.54	4	9.39	101.52	0.0743	not significant
Pure Error	0.0924	1	0.0924			
Cor Total	817.43	10				

Table 4.5 Fit statistics of the of Protein

Std. Dev.	2.74	**R^2**	0.954
Mean	50.41	**Adjusted R^2**	0.9079
C.V. %	5.44	**Predicted R^2**	0.4592
		Adeq Precision	13.3501

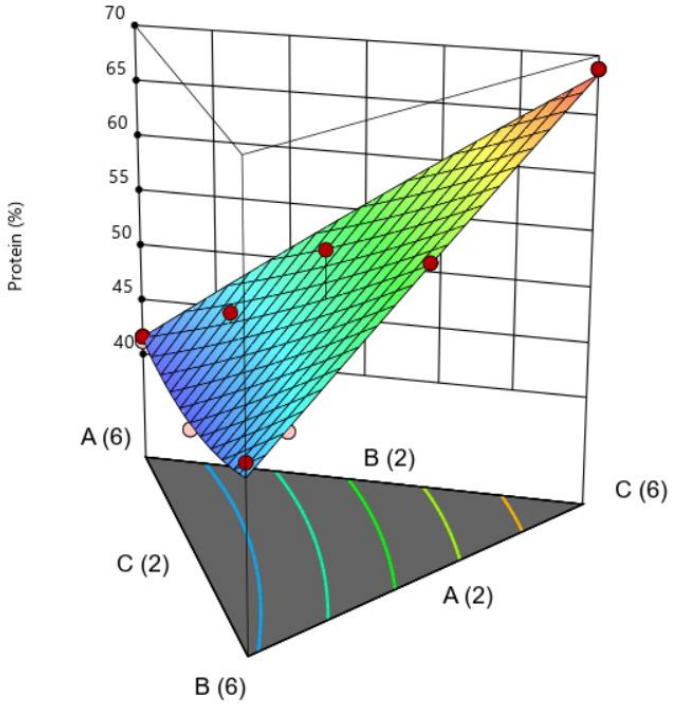


Fig. 4.2 Effects on composition of nutritional medium on Protein

The ANOVA for the linear model of Protein given in Table 4.4 showed that the model was significantly large with an F-value of 20.72 and a p-value of 0.0024, which reflected that the nutrient components significantly affected protein content in Spirulina. The linear term of the mixture was highly significant (F-value 51.58, p-value 0.0005), while the interaction terms AB, AC and BC were found to be not significant (p-values 0.5423, 0.8913 and 0.9576, respectively), indicating the dominance of the main effects. The lack-of-fit was not significant (p-value 0.0743), validating that the model well represented the experimental data. The fit statistics also helped validate the quality of the model with a high R^2 of 0.954, adjusted R^2 of 0.9079, predicted R^2 of 0.4592 and an adequate precision of 13.35, while the mean protein value was 50.41% and had a standard deviation of 2.74 and a low coefficient of variation (5.44%), which shows good precision and reliability of the model predictions for protein content given in the Table 4.5.

4.2 RESULT COMPOSITON OF THE OPTIMIZED NUTRITIONAL MEDIUM

Table 4.6 Solution obtained from the Response surface methodology

Number	Banana Peel Extract (ml)	Aquafaba (ml)	Peanut Meal Extract (ml)	Biomass (ml)	Protein (ml)	Desirability (ml)	
1	**2**	**2**	**6**	**9.605**	**68.388**	**0.982**	**Selected**
2	2	6	2	5.275	44.893	0.164	

The results of optimization by Response Surface Methodology provided the best combination of nutrients derived from food wastes for Spirulina growth. Of the conditions tested, the medium with 2 ml banana peel extract, 2 ml aquafaba and 6 ml peanut meal extract gave the highest performance with a biomass yield of 9.605 g, protein content of 68.388% and a desirability of 0.982 and was therefore the optimal solution chosen. Conversely, the blend of 2 ml banana peel extract, 6 ml aquafaba and 2 ml peanut meal extract produced merely 5.275 g biomass, 44.893% protein and a significantly lower desirability of 0.164, demonstrating the significance of proper nutrient balance as given in the Table 4.6. This demonstrates that the RSM model effectively predicted the best composition for achieving highest Spirulina biomass and protein content using low-cost, environmentally friendly media.

Table 4.7 Composition of media for the growth of the spirulina

Materials	Optimized composition
Banana peel extract	2ml
Aquafaba	2ml
Peanut meal extract	6ml

The best culture medium for Spirulina growth was optimized utilizing food waste-derived constituents obtained through Response Surface Methodology. The most effective

medium was a mixture of 2 ml banana peel extract, 2 ml aquafaba, and 6 ml peanut meal extract per 100 ml of water as given in the Table 4.7. This particular ratio delivered an optimal supply of carbohydrates, proteins and minerals to sustain maximum biomass productivity and protein content in the Spirulina grown, showing a successful, cost-effective and environmentally friendly substitute for standard synthetic media.

4.3 GROWTH OF SPIRULINA ON DIFFERENT NUTRITIONAL RATIOS

Preliminary trials were carried out to find the best nutrient medium-to-water ratio for Spirulina growth. Three dilutions were tested—1:10, 2:10, and 3:10—using the optimized medium prepared from banana peel extract, aquafaba, and peanut meal extract Figure 4.3. The results showed that the 1:10 ratio gave the highest biomass and protein yield, proving it to be the most suitable nutrient concentration for Spirulina growth. In comparison, the 2:10 and 3:10 ratios produced much lower yields, suggesting that excess nutrient concentration might actually reduce the growth. The bar chart clearly illustrates this trend, showing the rise and fall in biomass across the three ratios, and confirming that 1:10 is the ideal dilution for Spirulina culture under these conditions.

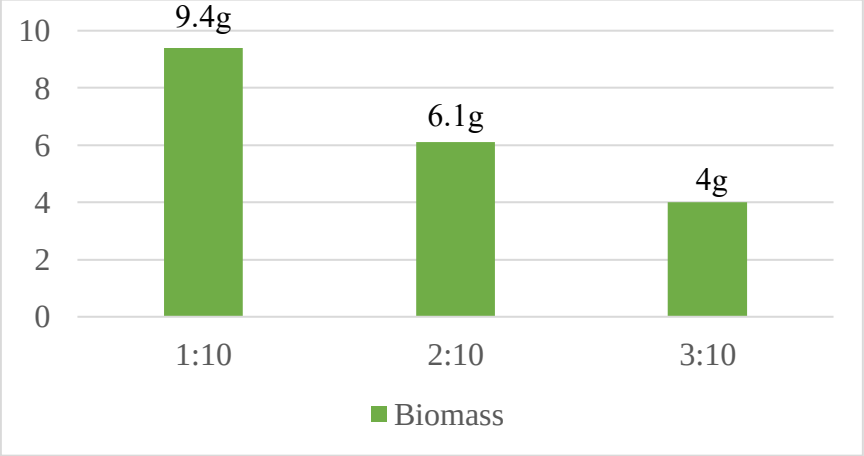


Fig. 4.3 Growth of spirulina on different nutritional ratios

4.4 ASH CONTENT

The ash content, which represents the total mineral content, was found to be 5.8% in the cultivated Spirulina and 6.4% in the commercial Spirulina Table 4.9. The slightly higher mineral content in the cultivated sample shows that even though it was grown on a low-cost, food waste-based medium under semi-outdoor terrace conditions, it still supported good mineral accumulation. This is likely because the waste-derived medium contained a variety of minerals from pulses, rice water, and other organic residues, which enriched the biomass. In comparison, commercial Spirulina grown under standardized conditions with optimized media showed a bit lower ash content, as its mineral inputs were more controlled. The higher ash content in the cultivated Spirulina indicates a richer micronutrient profile, making it suitable for use in functional foods and nutritional supplements. This result highlights that affordable, sustainable media can still produce Spirulina of high nutritional quality.

4.5 CARBOHYDRATES

The carbohydrate composition of the commercial Spirulina was 12.02%, whereas Spirulina grown in this study had 13.1% carbohydrates given in the Table 4.9. This difference could be attributed to differences in culture medium ingredients and environments. Commercial Spirulina is typically grown under highly standardized media with tightly controlled parameters, such that there are optimal carbon fixation and constrained carbohydrates. In contrast, experimental Spirulina was grown on the terrace under food waste water, where nutrient availability, intensity of sunlight, and pH levels could have accumulation of carbohydrate production. Even with the lower rate, the grown Spirulina yielded a high level of carbohydrate, indicating the viability of using food waste water as an environmentally friendly and inexpensive medium to produce carbohydrate-rich biomass.

4.6 WATER ACTIVITY

The commercially available Spirulina had a lower water activity of 0.537, which means it was drier and more shelf-stable, while the cultured Spirulina had a marginally higher water activity of 0.618 as given in the Table 4.9. The cultured sample was tray-dried, which could have led to slightly higher residual moisture than in the commercial process, performed under optimized industrial drying conditions. The increased water activity in the dried Spirulina is in the semi-controlled drying conditions and medium formulation but within a stable range for microbial stability. Appropriate drying allowed the biomass to maintain functional attributes with sufficient shelf life for food use.

4.7 PROTEIN

Protein composition in the commercial Spirulina sample was 72.05%, while Spirulina grown in this research had 68.25% protein content as given in the Table 4.9. The lower reading in the cultivated sample may be due to the application of food waste water as the culture medium and open terrace cultivation, where nutrient content, temperature, light intensity, and pH are less rigorously regulated compared to commercial factories. These are factors that can affect nitrogen uptake and biosynthesis of protein. However, the cultivated Spirulina nevertheless obtained a protein content that is high, meaning that food wastes water is enough to yield protein-enriched biomass that is on par with commercial quality but cost-efficient and environmentally friendly.

4.8 FAT CONTENT

The fat composition of the commercial Spirulina sample was 0%, whereas the Spirulina grown in this research contained 1.4% fat given in the Table 4.9. This is because using kitchen-waste-based media like banana peel extract, aquafaba and peanut meal extract, which

provide organic matter and micronutrients, enables them to cause the metabolic direction of Spirulina to change toward restricted lipid accumulation. Conversely, commercial Spirulina tends to be cultivated in highly controlled media designed for high protein content with low lipid synthesis. The semi-outdoor terrace environment—natural light, temperature variations, and relatively fluctuating concentrations of nutrients—could also have affected lipid biosynthesis. Consequently, the cultivated Spirulina had a minor but favorable fat component, which could add to its energy value and functional food characteristics over the commercial sample.

4.9 CHLOROPHYLL a

Commercial Spirulina has 12 mg/g (dry weight) of chlorophyll-a, whereas your cultured Spirulina (grown on terrace with food waste water medium) has 9 mg/g (dry weight) as shown in Table 4.9. The cultured Spirulina has less chlorophyll-a than the commercial one. This variation is due to the less favorable, fluctuating conditions in your experimental setup. Commercial Spirulina is commonly grown in tightly controlled light intensity, nutrient concentration (specifically nitrogen and trace micronutrients like iron and magnesium), pH, and temperature, all of which are ideal for optimal pigment production. In this case, use of food waste water from the as the medium on an open terrace exposes subjects' culture to varying light (shade phases, diel fluctuation), possible limitation or imbalances at times inferior pH fluctuation, and heat shock. All these stresses reduce the effectiveness of photosynthesis and biosynthesis of pigments, thereby reducing chlorophyll-a accumulation.

4.10 ß-CAROTENE

The β-carotene value in the commercial Spirulina is 1.1 mg/g dry wt., but in your experimental Spirulina (Terrace grown on food waste water media) it is 0.8 mg β-carotene/g dry wt. given in the Table 4.9. Your experimental Spirulina has a lesser amount of β-carotene than the commercial sample. This might be due to the less controlled cultivation conditions and fluctuation in nutrient supply in your experiment. Industrial Spirulina is usually cultivated under optimized light intensity, evenly balanced micronutrients (particularly elements such as iron, magnesium, and trace minerals), and steady pH and temperature—conditions that favor vigorous carotenoid production. Conversely, cultivation on an open rooftop using food waste water as media subjects the culture to variability in light, limitation of nutrients, and potential stress conditions, which decrease β-carotene biosynthesis. Nevertheless, obtaining a decent β-carotene concentration in your sample would demonstrate that even cost effective, waste-based media are capable of sustaining carotenoid accumulation, if not, maximal yield under the best commercial conditions.

4.11 MINERALS

The mineral content of the two Spirulina samples differed significantly. Cultivated Spirulina had sodium 240 mg, potassium 670 mg, calcium 0 mg, and lithium 0 mg, while commercial Spirulina had sodium 690 mg, potassium 450 mg, calcium 0 mg, and lithium 0 mg given in the Table 4.9. This suggests that the industrial sample contained more sodium, possibly because the controlled growth medium in industrial production, and the natural Spirulina, which was cultivated on a medium based on food waste, contained more potassium, which approximates the natural mineral content of organic substrates employed. Neither sample contained calcium or lithium. The differences underscore how the growth medium composition has direct effects on the mineral profile of Spirulina, whereby the harvest biomass continues to retain substantial levels of key minerals such as potassium, rendering it fit for use in functional food and nutritional products.

4.12 COLOUR

The commercial Spirulina had a deeper green color, with a smaller L* value (52.4) showing a darker, more intense pigment, and more negative a* values, demonstrating more intense green strength. The cultivated Spirulina under a medium made of food waste had a marginally greater L* value (49.8), showing lighter, brighter green given in the Table 4.9. These variations are mostly because of the conditions of cultivation: commercial Spirulina is cultivated under tightly controlled parameters and optimum light, temperature, and nutrients with maximum chlorophyll-a and phycocyanin synthesis. Semi-outdoor terrace cultivation in low-cost medium resulted in slight deviations from light intensity, temperature, and nutrient levels with a decrease in pigment concentration. In spite of this, the production Spirulina retained a typical green color, verifying that the waste-based medium supported functional pigment production amenable to food uses successfully.

4.13 ANTIOXIDANT

Antioxidant activity of the commercial Spirulina was 47.07%, while cultivated Spirulina grown have 41.33% given in the Table 4.9. This decline in antioxidant capacity in the cultured sample is due to the semi-outdoor terrace conditions and the utilization of kitchen-waste-based media (banana peel extract, aquafaba and peanut meal extract), which could be less effective in delivering the same micronutrient and pigment background as commercial media that are optimized for bioactive compound production. Variables like changes in light intensity, temperature and pH, and lack of some trace elements (e.g. iron, zinc) can lower the production of antioxidant pigments like phycocyanin, chlorophyll and carotenoids. However, the cultivated Spirulina from the cost effective, eco-friendly food waste media still possessed a high antioxidant activity, proving that inexpensive, environmentally friendly food waste media can yield biomass with great functional attributes as good as commercial Spirulina.

4.14 ENERGY VALUE

The energy value in Spirulina samples varied based on macronutrient differences. The commercial Spirulina contained a greater energy value of 400 kcal, which may indicate its maximized production environment and potentially greater carbohydrate and protein levels. Cultivated Spirulina, however, grown with a medium prepared from food waste and semi-outdoor terrace culture, possessed a lower energy value of 345 kcal given in the Table 4.9. This variation can be explained by differences in nutrient availability and biomass composition during non-industrial growth conditions. Although the caloric value is lower, the cultivated Spirulina remains a nutrient-rich source of protein, minerals, and bioactive compounds that can find functional food applications while enabling sustainable low-cost production.

4.15 CRUDE FIBRE

The crude fiber of the commercial Spirulina sample was 5.2%, whereas that of the Spirulina grown in this research was 4.8%. The minor reduction in fiber in the grown sample can be explained on the basis of variations in the nutrient composition of the food waste-medium and the open terrace growth conditions. Commercial Spirulina is generally cultured under optimal nutrient and aeration that favor even cell wall development, resulting in higher fibre deposition. The experimental medium, despite high natural nutrient levels, could have provided inconsistent carbon and mineral availability, resulting in lower fiber deposition. Nevertheless, the variation is small, showing that the cultured Spirulina had a similar crude fiber content that is suitable for dietary and functional food uses.

4.16 MOISTURE CONTENT

The moisture content of the commercial Spirulina sample was 4.1%, while the Spirulina grown in the present study had a slightly higher value of 5.4%. The rise in the moisture

content in the cultivated spirulina can be attributed to difference in the drying conditions, humidity levels, and the nature of the composition of the food waste-based medium. Because the cultivation was performed on the terrace in semi-controlled conditions, total dehydration may not have been attained than from the commercial sample, which is industrially dried with controlled temperature and air flow. The difference in moisture content is still within acceptable limits, which shows that the Spirulina grown was effectively dried and could be used for further processing and storage.

4.17 COST ESTIMATION

Table 4.8 Cost estimation

Cost Component	Commercial Spirulina (₹)	Cultivated Spirulina (₹)	Remarks
Raw Materials (media)	60	20	Commercial uses synthetic nutrients; cultivated uses food waste (minimal cost).
Culture Setup & Maintenance	40	30	Includes beakers, pH adjustment, terrace space, etc.
Electricity (drying)	25	15	Tray drying at 40–50°C; no large-scale machinery.
Labor & Handling	30	30	Manual handling and terrace-based cultivation reduce costs.
Packaging & Storage	20	20	Commercial requires standardized packaging; cultivated uses simple storage.
Overheads (transport, QC)	35	35	Commercial includes distribution & quality certification.
Total Cost (per 100 g)	**₹210**	**₹150**	Cultivated Spirulina ~28.5% cheaper.

Table 4.9 Comparative analysis of Cultivated and Commercial Spirulina

S.No	Analysis	Commercial Spirulina	Cultivated Spirulina
1.	Carbohydrates	12.02%	13.1 %
2.	Protein	72.05%	68.2%
3.	Fat	0%	1.4%
4.	Crude fiber	5.2%	4.8%
5.	Moisture	4.1%	5.4%
6.	Ash	5.8%	6.4%
7.	Sodium	690mg	340mg
8.	Potassium	250mg	670mg
9	Lithium	0	0
10.	Calcium	0	0
11.	Antioxidants	47.07%	41.33%
12.	Color	52.4	49.8
13.	Water activity	0.537	0.618
14	Chlorophyll A	12 mg	9 mg
15.	Beta carotene	1.1 mg	0.8 mg
16.	Energy value	400 kcal	345 kcal

CHAPTER 5

CONCLUSION

This project effectively proved cultivation of Spirulina on low-cost, cost-effective food waste-based nutrient sources as a viable green alternative to traditional synthetic media. Banana peel extract, aquafaba and peanut meal extract were blended and optimized by Response Surface Methodology (RSM) to obtain the best combination for biomass yield and protein enrichment. The mother culture was obtained from Viridia Biotech Company, Karur and the culture was done on the terrace of Kongu Engineering College under natural sunlight and 30–38 °C ambient temperature with an alkaline medium (pH 8.5–10). This outdoor system showed visible biomass development within 8–10 days and the harvested Spirulina was filtered, tray-dried and pelletized, validating the feasibility of an ecologically friendly and low-cost process of production.

Nutritional examination of the Spirulina harvested indicated that it had 13.1 % carbohydrates, 68.2 % protein and 1.4 % fat whereas commercial Spirulina had 12.2 % carbohydrates, 72.05 % protein and 0 % fat. The slightly lower percentage of carbohydrates and proteins in the cultivated sample was probably due to naturally occurring fluctuations in light intensity, temperature and nutrient content in terrace-based cultivation and food-waste media. In contrast, the occurrence of a small but positive lipid occurrence in the cultured Spirulina indicates the impact of organic matter and micronutrients in the alternative medium on the metabolic pathways, favoring restricted lipid synthesis. Apart from these variations, the cultured Spirulina retained deep green color and high content of pigments, indicating healthy photosynthetic activity.

Overall, this study demonstrates that nutrient-rich food waste can be effectively used to grow high-quality Spirulina under simple semi-outdoor conditions. This approach not only reduces production costs but also promotes waste utilization and supports economical reduction, while producing biomass and protein with nutritional quality comparable to commercial standards. By combining low-cost media with optimization tools like RSM, the study presents a cost-effective and sustainable model for Spirulina cultivation, which can support the development of functional foods and environmentally friendly microalgal production.

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